

Experiment 17: Edge Detection using Sobel Matrix along X-axis

AIM

To detect horizontal edges using the Sobel X operator.

PROCEDURE

1. Read the image.
2. Convert it to grayscale.
3. Apply Sobel operator along X-axis.
4. Display the result.

PROGRAM

```
import cv2

img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
cv2.imshow("Original Image", img)
cv2.imshow("Sobel X", sobelx)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT



Explanation

We applied the Sobel operator along the X-axis. It detects intensity changes in the horizontal direction, highlighting vertical object edges.